Space and Beyond logo

SWASTHYAGRID

THEME: SMART WORLD, BETTER FUTURE

Moves the medicine a district already owns, before the state buys more

OUR IDEA: BEFORE INDIA BUYS MORE MEDICINE, MOVE WHAT EACH DISTRICT ALREADY OWNS. ONE MORNING VIEW, ONE RANKED TRANSFER, ONE OFFICER'S SIGN-OFF.

An estimated 1.36 crore stock-out days avoided a year across India's 38,241 PHCs and CHCs in the base case, from a stated model. This deck presents the idea; the prototype is shown separately in Round 2.

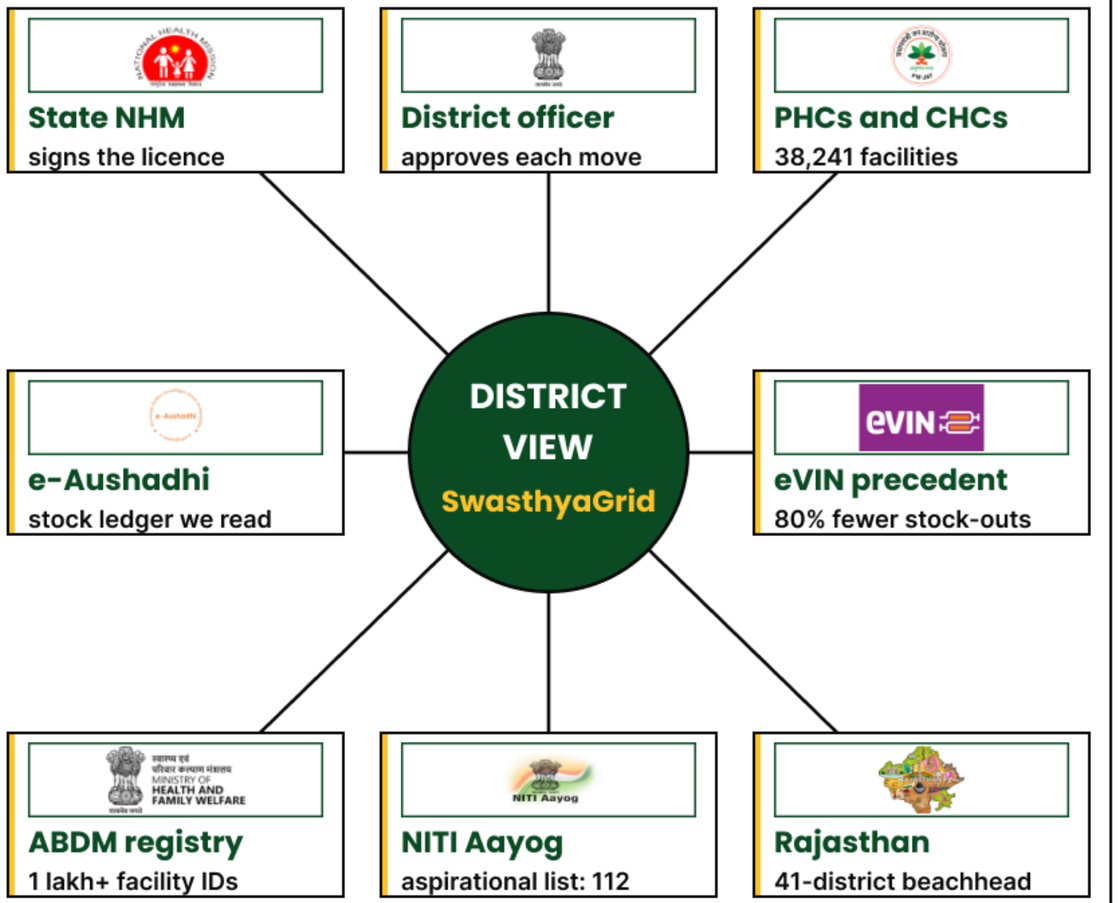
TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University (DTU)	Team Leader
Gursimran Kaur	Guru Gobind Singh Indraprastha University	Member

harshkawatra11@gmail.com | file: TeamIdeaForge_SpaceAndBeyond

ONE DISTRICT VIEW, EIGHT ACTORS IT CONNECTS

₹558 Cr of medicine expired in two audited states while PHC shelves ran short: the fix sits inside each district.



Logos mark public bodies referenced in the analysis; no partnership or endorsement is implied.

Source: research sections 2, 6, 7, 8 | CAG audits (Telangana 2024, Odisha 2026) | UNDP eVIN | MoHFW HDI 2022-23 | NITI Aayog | Team IdeaForge model (NEXORA model, unchanged)

Districts run short and write off expired medicine at once; moving what a district already owns avoids an estimated 1.36 crore stock-out days a year

THE PARADOX

SHORT AND WASTED AT ONCE

66 to 80% key-medicine availability at PHCs in two state studies

₹558 Cr of medicine expired in Telangana and Odisha (CAG)

79.9% of CHC specialist posts unfilled, 4,413 of 21,964 in place

43.4% of health spending paid out of pocket, 2022-23

WHY THE OBVIOUS FIXES STALL

Obvious fix	Why it stalls
Hire more	79.9% of CHC specialist posts already empty
Buy more	₹558 Cr already expired in two audited states
One database	health is a State subject (List II, Entry 6)
New dashboard	five systems record stock; none moves it

So what: none uses what a district already holds.

FIVE SYSTEMS RECORD, NONE MOVES

System	Records	Medicine	Moves
HMIS	Part	Part	No
e-Aushadhi	Yes	Yes	No
eVIN (vaccines)	Yes	No	No
IHIP / IDSP	No	No	No
ABDM registry	No	No	No
SwasthyaGrid	Yes	Yes	Yes

THE ANSWER

MOVE FIRST, BUY ONLY FOR THE GAPS

1 SEE
Each morning, any item under 3 days of cover is flagged; 5 days for emergency drugs

2 PROPOSE
The nearest surplus within 40 km is ranked and sized to refill 21 days of cover

3 APPROVE
A named officer approves, edits or rejects; the receiving PHC confirms delivery

THREE REASONS EACH SIDE

Reason	Against buy or hire	For move first
Numbers	79.9% of specialist posts already empty	up to 1 in 3 key medicines missing on a day
Fit	buying feeds ₹558 Cr of audited expiry	eVIN cut vaccine stock-outs by 80%
Delivery	a national database clashes with the State List	reads exports states already produce

THE PRECEDENT: eVIN

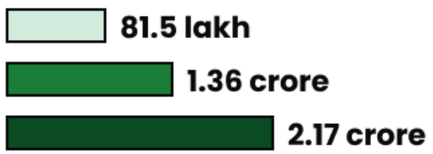
80% fewer stock-outs

29,000+ cold-chain points

12 to 36 states and UTs

THE PAYOFF

THREE SCENARIOS, DAYS AVOIDED



Base case assumes half of eVIN's 80% cut; availability rises from 72.2% to 86.1%

WHAT A DISTRICT PAYS, AND GETS BACK

₹12 L licence a district a year

0.25% of the district's central NHM share

6.1 to 12.5% of yearly expiry rescued repays it

10.65x lifetime value to acquisition cost

WHERE EACH ASK IS ANSWERED

Ask in the brief	Where on the deck	Slide
Problem	shelf, waste, root cause	3
Solution, innovation	loop, rules, system grid	4
Market, business model	sizing, buyer, price	5
Implementation	six-week plan, risks	6
Impact, future scope	model, horizons	7
Assumptions, sources	workings, register	8

Source: MoHFW HDI 2022-23 | PMC8654139 | CAG via Medical Dialogues and UNI | NHA 2022-23 | UNDP eVIN | PRS DFG 2025-26 | research sections 2, 7, 9, 11, 14 | model

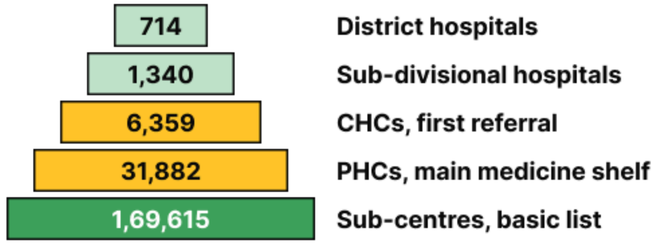
A third of key medicines can be missing from a PHC shelf while state stores write off crores in expired stock: no one sees a district's supply as one

THE SHELF

AVAILABILITY OF KEY MEDICINES, FOUR STUDIES, %

Range: 45% to 80% by state; up to 1 in 3 items missing

38,241 PHCs AND CHCs ARE THE SHELF



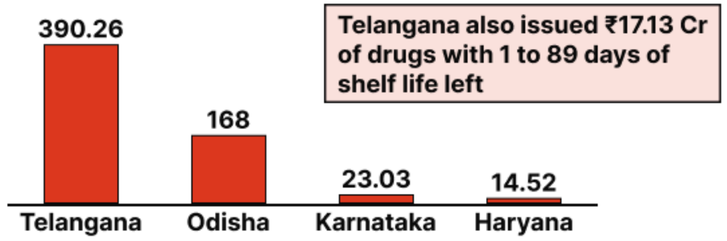
WHERE A MISSING DRUG COSTS MOST

Condition	Tracer drug	Burden in India
Snakebite	Antivenom	58,000 deaths a year (eLife)
Rabies	Anti-rabies	5,726 deaths a year (WHO)
Childbirth	Oxytocin	MMR 93 per 1,00,000 births
Dengue	Fluids	signal 4 to 6 weeks pre-peak

Duration: 60% of Haryana's gaps last 3 to 6 months

THE WASTE

CAG: EXPIRED IN STATE STORES, ₹ CR



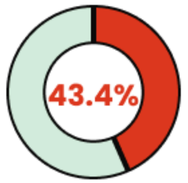
Telangana also issued ₹17.13 Cr of drugs with 1 to 89 days of shelf life left

PER DISTRICT, EVERY YEAR

Step	Telangana	Odisha
Expired (audited)	₹390.26 Cr	₹168.00 Cr
Years audited	6.00	5.83
Per year	₹65.04 Cr	₹28.80 Cr
Districts	33	30
Per district a year	₹1.97 Cr	₹0.96 Cr

So what: ₹12 L repays at 6.1% to 12.5% of this

WHO PAYS WHEN THE SHELF IS EMPTY



Measure	Value
Out of pocket, 2013-14	64.2%
Out of pocket, 2022-23	43.4%
Pharmacy share of CHE	19.35% to 21.19%
Buy privately, Odisha	over 70%
CHC specialists missing	79.9%

ROOT CAUSE

FIVE BRANCHES, AND WHAT A DISTRICT VIEW FIXES

Why is a tracer drug missing when the officer checks the shelf

1. Supply undersized or late at the source	Not fixed
2. Supply exists, but in the wrong facility	Fixed
3. Records lag the physical shelf by days	Partly
4. Demand spikes the plan did not expect	Partly
5. Nobody with authority sees it in time	Fixed

FIVE WHYS ON ONE STOCK-OUT

WHY 1 ARV ran out: three bites in a week drained the stock → **WHY 2** The register is paper, checked when there is time → **WHY 3** The store cannot see which of 49 PHCs runs low → **WHY 4** Facility records are never added up district-wide → **WHY 5** Adding them up each morning is nobody's job

Root cause: the medicine is often already inside the district; no one is tasked to see it.

Source: MoHFW HDI 2022-23 | PMC8654139, PMC10292169, Kotwani 2015 | eLife 2020 | WHO India | SRS 2019-21 | CAG via Medical Dialogues, UNI, Deccan Herald, Tribune | NHA 2022-23 | PMC11836861

SwasthyaGrid adds what five public systems lack: a morning view of every shelf, the nearest surplus ranked, and an officer's sign-off on every move

THE LOOP

THE DAILY LOOP: SEE, FLAG, PROPOSE, APPROVE, CONFIRM

1 SEE	2 FLAG	3 PROPOSE	4 APPROVE	5 CONFIRM
District officer one view of stock, beds and staff in every facility	No person any item under 3 days of cover is marked	No person nearest surplus ranked; move sized to 21 days	Named officer approve, edit or reject; name and time logged	Receiving PHC delivery confirmed; stock updates once

Every confirmed delivery feeds the next morning's view

THE RULES, IN AN OFFICER'S WORDS

Rule	Value
Days of cover	units in stock ÷ average daily use
Flag, routine items	under 3 days of cover
Flag, emergency drugs	under 5 days (antivenom, anti-rabies)
Refill target	21 days of cover
Transfer radius	within 40 km
Bed redirect	next case to a facility under 90% full
Patient data	none, by design

HOW IT DECIDES

FOUR OUTCOMES FOR EVERY FLAG

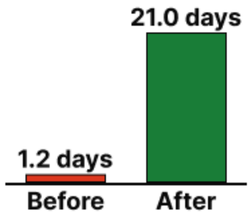
Refill from buffer
district store tops the PHC up to 21 days

Lateral transfer
a PHC within 40 km sends its surplus

Bed redirect
next case goes to a facility under 90% full

Escalate
no surplus anywhere: flagged for purchase

ONE TRANSFER, WORKED THROUGH



1. **See:** 7 ARV vials, 1.2 days of cover
2. **Flag:** under the 3-day line
3. **Propose:** 119 vials, district store
4. **Approve:** officer signs, no edit
5. **Confirm:** 126 vials, 21.0 days

Worked example on simulated data

BEFORE AND AFTER, FOR THE DISTRICT OFFICER

Task	Today	With SwasthyaGrid
Find a shortage	by phone, too late	flagged that morning
Pick a source	calls 2 or 3 PHCs	one ranked proposal
Record it	nothing kept	name, time logged
Expiring stock	written off later	flagged weeks early
Daily decisions	one per call	one per flagged item

WHY IT IS NEW

FIVE PUBLIC SYSTEMS RECORD; NONE PROPOSES A MOVE

System	Stock	District	Medicine	Proposes	Sign-off
HMIS	Part	No	Part	No	No
e-Aushadhi	Yes	Part	Yes	No	No
eVIN	Yes	Yes	No	No	No
IHIP / IDSP	No	Part	No	No	No
ABDM registry	No	Part	No	No	No
SwasthyaGrid	Yes	Yes	Yes	Yes	Yes

ERRC: WHAT CHANGES FOR THE OFFICER

ELIMINATE
phone calls to find surplus

REDUCE
one approval per flagged item

RAISE
speed from flag to sized move

CREATE
a move proposed across facilities

CONCLUSIVELY

The medicine is often already inside the district. The missing piece is a daily, ranked proposal a named officer signs, with no patient data and no new hardware.

Officer signs | No patient data | Data stays home

Source: research section 8 (design rules; worked example on simulated data) | C-DAC e-Aushadhi | UNDP eVIN | ABDM HFR | IHIP | Kim and Mauborgne (2005) | research section 7 capability matrix

State health departments buy for about 780 districts, a ₹93.6 Cr licence pool; we enter through 112 aspirational districts and Rajasthan's 41

SIZE

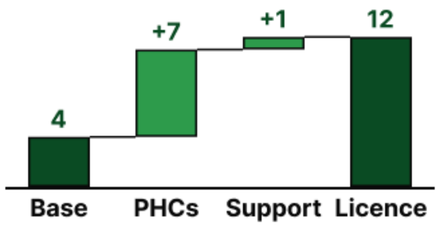
BOTTOM-UP: DISTRICTS × ₹12 L LICENCE

TAM 780 ₹93.6 Cr
SAM 112 ₹13.44 Cr
SOM 80 ₹9.6 Cr
Rajasthan 41 ₹4.92 Cr

Segment	Basis
TAM	LGD, about 780
SAM	NITI list, 112
SOM	base path, Y3
Entry	Rajasthan, 41

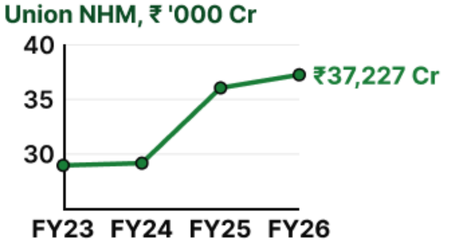
SOM: 10.3% of TAM. Rajasthan: 5 of 41 aspirational

THE ₹12 L LICENCE, BUILT UP, ₹ L



- ₹4 L base platform
- ₹7 L = 50 × ₹14,000
- 49.0 facilities per district

TOP-DOWN CHECK: 0.25% OF NHM SHARE



₹47.73 Cr
central NHM share per district

0.25%
of it pays the licence

BUYER

WHO DECIDES: THE BUYING CENTRE

Role	Who	What they need
Signs	State NHM director	an NHM budget line
Champions	District Collector	health weighs 30%
Uses daily	DHO	one approval per flag
Reports	Pharmacist, MO	one confirmation
Benefits	Patients	drug on the shelf

STP: SEGMENT, TARGET, POSITION

Step	Our choice
Segment	about 780 districts, split by a scored health goal and a digital stock export
Target	112 aspirational districts first; Rajasthan's 41 as the beachhead
Position	not a new system to learn: a layer that moves what a district owns

Rajasthan: the design models 5 of its districts

WHY NOW: FOUR RAILS ALREADY EXIST

95.99 Cr
ABHA IDs created by 31 Jul 2026

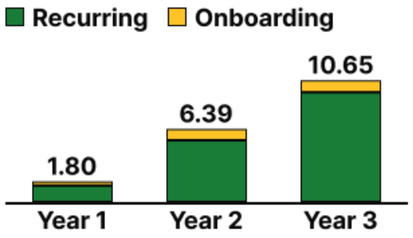
31
States and UTs on the e-Aushadhi dashboard

1 lakh+
facilities with an ABDM registry ID

₹37,227 Cr
NHM budget 2025-26, up every year

MONEY

REVENUE PATH, ₹ CR, BASE CASE



Case	Districts	ARR Y3
Bear	50	₹6.0 Cr
Base	80	₹9.6 Cr
Bull	105	₹12.6 Cr

Year 3: 90.1% of billings recur; ₹3 L onboarding per new district

UNIT ECONOMICS, ONE DISTRICT, BASE

71% gross margin
₹4 L cost to win
5.6 mo payback
10.65x LTV to CAC

LTV: ₹12 L × 71% × 5 years = ₹42.6 L, against ₹4 L to win the district

PRICE AGAINST INDIAN HEALTH SOFTWARE

₹12 L a district

₹3 L, entry HMIS — ₹25 L, enterprise EHR

A vendor-price scan, not a signed price

Source: LGD (about 780 districts) | NITI Aayog list and ADP 2018 | Scroll (Rajasthan 41) | PRS DFG 2024-25, 2025-26 | PIB ABDM Jul 2026 | C-DAC | Codingclave, upGrowth 2026 | model

A district can go live six weeks after a data MoU, on exports the state already produces: three feasibility tests are green and two stay amber

CAN IT RUN

TELOS: THREE GREEN, TWO AMBER

Test	Evidence	Rating
Technical	reads exports states already run	GREEN
Economic	licence is 0.25% of NHM share	GREEN
Legal	no patient data in scope	GREEN
Operational	needs a daily officer habit	AMBER
Schedule	MoU must land in week 1	AMBER

RESOURCES REQUIRED: NO NEW HARDWARE

Resource	What it takes
Data	nightly stock export the state makes
Facility IDs	ABDM registry IDs, 1 lakh+ facilities
People	nodal officer; DHO signs off daily
Devices	phones and PCs in use; offline entry
Money	₹12 L licence, ₹3 L onboarding
Team	first hire paid by first fee

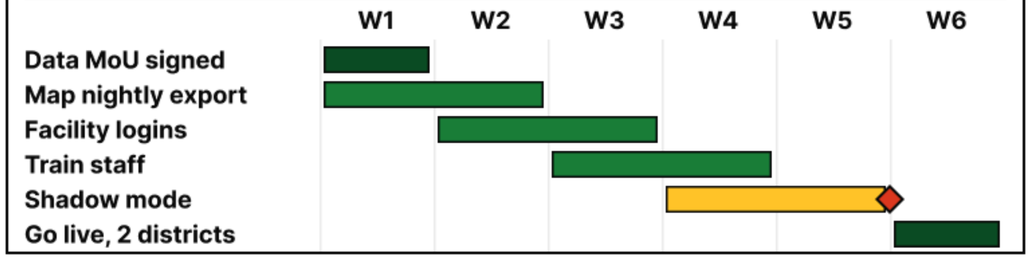
WHY A STATE CAN SAY YES: THE LAW

Instrument	What it means here
State List	each state licenses; data stays home
DPDP Act, 2023	no patient data in scope
CDSCO 2025	governs stock that could not move
GeM	a standing route to buy software

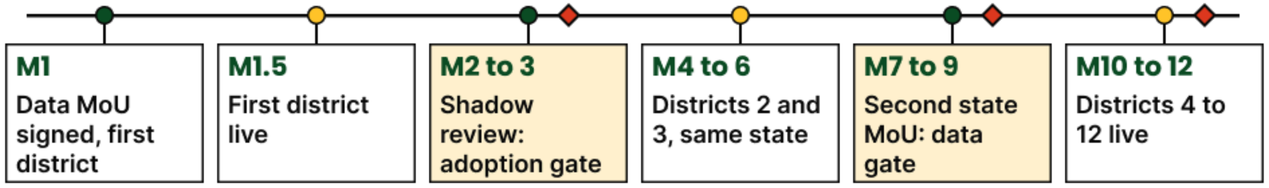
Our reading: counsel confirms before any contract

THE PLAN

SIX WEEKS FROM MoU TO A LIVE DISTRICT



TWELVE MONTHS: FROM 1 TO 12 DISTRICTS



RISKS

SIX RISKS, EACH WITH A FIX

	I low	I med	I high
P high	R5	R1	
P med		R3 R6	R2
P low			R4

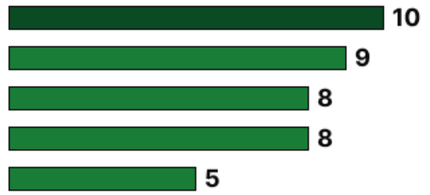
- R1 freshness flag on data
- R2 buffer when use rises
- R3 alert on an idle queue
- R4 one keen district first
- R5 offline entry, sync
- R6 hire from pilot fees

THREE STOP RULES, WRITTEN IN ADVANCE

Rule	Stop if
Adoption	under 50% approved after 4 weeks
Data	no MoU in 8 weeks, in two states
Expiry	pilot expiry share under 6.1%

Funding: no investment asked; a paid pilot, or an AIM digital-health grant of up to ₹65 lakh

40 INTERVIEWS PLANNED, NONE DONE



A plan, not a result: none done yet

Source: research sections 13 and 15 | Constitution, List II, Entry 6 | DPDP Act 2023 | CDSCO May 2025 | ABDM HFR | NITI Delta Ranking | AIM INDovation | Appendix E validation plan

Across 38,241 PHCs and CHCs, SwasthyaGrid would avoid an estimated 1.36 crore stock-out days a year, lifting availability from 72.2% to 86.1%

THE NUMBER

STOCK-OUT DAYS AVOIDED A YEAR, BASE CASE

1.36 crore

72.2% to 86.1%	17,412	355
availability	a district a year	a facility a year

THE MODEL, ONE LINE AT A TIME

#	Step	Value
1	PHCs + CHCs	38,241
2	× tracer medicines	7
3	× days in a year	365
4	× share of days out (1 − 72.2%)	27.8%
5	= baseline stock-out days	2.72 crore
6	× share removed (base)	50%
7	= days avoided a year	1.36 crore

SENSITIVITY: EVERY CELL CLEARS 58.6 LAKH

Availability	30% cut	50% cut	80% cut
66% today	99.7	166.1	265.8
72.2% (mid)	81.5	135.8	217.3
80% today	58.6	97.7	156.3

Lakh stock-out days avoided a year; base cell 135.8

HOW IT HAPPENS

THEORY OF CHANGE: FROM A DATA READ TO A STOCKED SHELF

INPUTS	ACTIVITIES	OUTPUTS	OUTCOMES	IMPACT
daily stock, bed and staff data states hold	flag, propose, approve, move, confirm	transfers approved the day they are proposed	fewer stock-out days and less expiry	emergency drugs on the shelf when needed

THREE SCENARIOS AND WHAT EACH ONE ASSUMES

Case	Days avoided	Availability	Why this cut
Bear, 30%	81.5 lakh	80.5%	a floor well below any reported result
Base, 50%	1.36 crore	86.1%	half of eVIN's reported 80% cut in stock-outs
Bull, 80%	2.17 crore	94.4%	eVIN's own result, used as a ceiling

Per district: 17,412 days a year across about 49 facilities; per facility, 355 days

Limit: a logistics estimate for 7 tracer medicines, not a count of lives saved

FUTURE SCOPE

PEOPLE, PLANET, PROFIT

PEOPLE	PLANET	PROFIT
43.4% health spend paid out of pocket	**4.4%** of global emissions, 71% supply chain	**71%** gross margin, base case
58,000 snakebite deaths	**82.1%** bin unused drugs	**10.65x** LTV to CAC

FUTURE SCOPE: THREE HORIZONS

H1 CORE	H2 EMERGING	H3 FUTURE
0 to 12 months 12 districts, ₹1.44 Cr ARR	12 to 36 months 80 districts, base ₹9.6 Cr ARR	36 months on about 780 districts, not projected

Gate: each horizon opens only when the last one's gate is met

BEYOND YEAR 3: THE SAME LAYER, WIDER

- **Wider basket:** beyond 7 tracer drugs to full lists
- **Blocks:** 500 aspirational blocks since Jan 2023
- **Same buyer:** nutrition and agriculture, later

SDG 3.8	SDG 10.2	SDG 12.3

A contribution to each target, not a measured change

Source: NEXORA model (impact, unit economics) | MoHFW HDI 2022-23 | PMC8654139 | UNDP eVIN | NHA 2022-23 | eLife 2020 | HCWH and Arup 2019 | PMC10867696 | WHO | UN SDG metadata

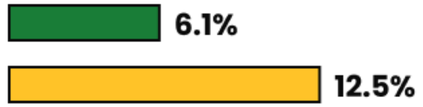
Every number traces to a named source or one stated model; even the weakest mix of assumptions still avoids 58.6 lakh stock-out days a year

A1 MODEL

A1 MARKET AND UNIT ECONOMICS WORKINGS

Line	Working
TAM	780 × ₹12 L = ₹93.60 Cr
SAM	112 × ₹12 L = ₹13.44 Cr
SOM, Y3 base	80 × ₹12 L = ₹9.60 Cr
Beachhead	41 × ₹12 L = ₹4.92 Cr
Gross profit a month	₹12 L × 71% ÷ 12 = ₹0.71 L
CAC payback	₹4 L ÷ ₹0.71 L = 5.6 months
LTV : CAC	₹42.6 L ÷ ₹4 L = 10.65x

A1 CUSTOMER PAYBACK FROM AUDITED EXPIRY



Share = ₹12 L ÷ (audited expiry ÷ years ÷ districts)

A1 REVENUE PATH, ₹ CR, THREE CASES

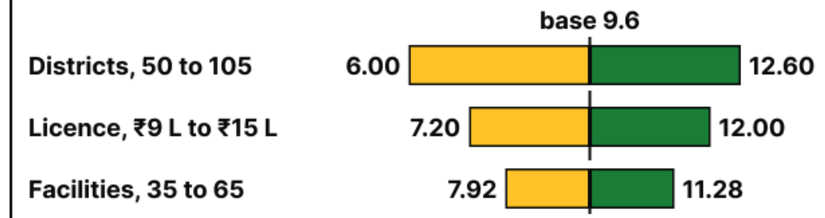
Line	Y1	Y2	Y3	Bill Y3	GM
Bear ARR	1.44	3.60	6.00	6.60	62%
Base ARR	1.44	5.40	9.60	10.65	71%
Bull ARR	1.44	7.20	12.60	13.95	76%
Bear districts	12	30	50		
Base districts	12	45	80		
Bull districts	12	60	105		

A2, A3

A2 THE ASSUMPTIONS THAT MOVE THE ANSWER

ID	Assumption	Value	Basis, and how a pilot tests it
A01	Survey-day availability proxy	72.2%	standard proxy; replaced by daily logs
A02	Stock-out days removed	30, 50, 80%	eVIN reported 80%; base is half
A05	Licence a district a year	₹12 L	₹4 L + ₹7 L + ₹1 L; first signed price
A08	Gross margin	62, 71, 76%	Indian SaaS benchmarks; real delivery cost
A09	Cost to win a district	₹4 L	team estimate; cost of the first close
A12	Districts at year end	12, 45, 80	base path; pace of sign-ups
A16	MoU to a live district	6 weeks	a plan; the first district's real time

A3 YEAR-3 ARR SENSITIVITY, ₹ CR



Reading: the pace of district signing moves Y3 ARR most

A4 SOURCES

A4 SOURCES THE DECK LEANS ON

S01 MoHFW, Health Dynamics of India 2022-23
S02 PMC8654139, Puducherry availability study
S03 PMC10292169, Gadag Taluk availability
S44 Kotwani et al. 2015, Punjab and Haryana
S06 Suraweera et al. 2020, eLife, snakebite
S10 MoHFW, National Health Accounts 2022-23
S12 UNDP India, eVIN programme page
S13 Constitution, Seventh Schedule, List II
S14 Digital Personal Data Protection Act 2023
S15 C-DAC, e-Aushadhi (DVDMS)
S16 CAG Telangana via Medical Dialogues 2024
S17 CAG Odisha via UNI, Oct 2026
S18 PRS, Demand for Grants 2025-26, Health
S19, S20 NITI Aayog, aspirational districts
S21 PIB, ABDM statistics, Jul 2026
S25, S26 Codingclave, upGrowth, 2026

A4 SHORT FORMS USED ON THIS DECK

Short	Meaning	Short	Meaning
PHC	primary health centre	CHC	community health centre
NHM	National Health Mission	DHO	district health officer
CAG	Comptroller and Auditor General	OOP	out-of-pocket spending
ABDM	Ayushman Bharat Digital Mission	ABHA	Ayushman Bharat Health Account
eVIN	electronic vaccine network	DVDMS	drug and vaccine ledger
ARR	annual recurring revenue	CAC	cost to win a customer
MoU	memorandum of understanding	ARV	anti-rabies vaccine

THANK YOU

THEREFORE

Before India buys more medicine, it should move what each district already owns: one morning view, one ranked transfer, one officer's sign-off, and an estimated 1.36 crore stock-out days avoided a year in the base case.

1.36 crore days a year, an estimate | **₹12 L a district, 0.25% of its NHM share** | **6 weeks from MoU to a live district**

TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University (DTU)	Team Leader
Gursimran Kaur	Guru Gobind Singh Indraprastha University	Member

Space and Beyond logo

The prototype and its user flow are demonstrated in Round 2

harshkawatra11@gmail.com | Team IdeaForge | Space & Beyond 2026, NSUT

Source: Team IdeaForge, Space & Beyond Innovation Ideathon 2026, NSUT | research and model: swasthyagrid-nexora-research